

TLS Trainer — Backup & Recovery System: Test Report

Date: 2026-05-27 Environment: Dev (localhost:4200)

Summary

Test	Description	Result
T1	Create manual backup	☒ PASS
T2	List backups	☒ PASS
T3	Stats endpoint (empty)	☒ PASS
T4	Stats after backup created	☒ PASS
T5	Download backup as JSON	☒ PASS
T6	Bulk JSON export	☒ PASS
T7	SQL export (.sql dump)	☒ PASS
T8	Restore from backup	☒ PASS
T9	Pre-restore auto-snapshot created	☒ PASS
T10	Delete backup entry	☒ PASS
T11	Daily auto-backup scheduler	☒ PASS
T12	Weekly auto-backup scheduler	☒ PASS

12/12 PASS — 0 FAIL

Test Details

T1 — Create Manual Backup

```
POST /api/admin/backup/create {"note":"test backup run"}
Response: {"ok":true, "id":"manual-1779923891917-6m1ld", "sizeBytes":48690}
Tables captured: 20 tables including trainees(3), modules(9), questions(25), achievements
```

T2 — List Backups

```
GET /api/admin/backup/list
Returned 1 entry with correct id, label, note, created_at, size_bytes, table_counts
```

T3 + T4 — Stats Endpoint

```
Before backup: {totalBackups:0, totalBytes:0, lastBackup:null}
After backup:  {totalBackups:1, totalBytes:48690, lastBackup:{label:"manual",...}}
                counts: {manual:1, daily:0, weekly:0, pre-restore:0}
```

T5 — Download Backup as JSON

```
GET /api/admin/backup/:id/download
Headers: Content-Disposition: attachment; filename="TLS-backup-manual-2026-05-27-6m1ld.js"
Body keys: {meta, data} – meta contains id/label/note/version/timestamps
Data: all 20 tables present, trainees count = 3 ✓
```

T6 — Bulk JSON Export

```
GET /api/admin/backup/export/json
Content-Disposition: attachment; filename="TLS-export-2026-05-27.json"
All 20 tables present ✓
```

T7 — SQL Export

```
GET /api/admin/backup/export/sql
Content-Disposition: attachment; filename="TLS-export-2026-05-27.sql"
Output: proper SQL with DELETE + INSERT OR REPLACE per table
Header comment includes export timestamp and version ✓
```

T8 — Restore

```
POST /api/admin/backup/:id/restore
Response: {"ok":true, "tablesRestored":16}
(4 tables skipped: backups itself + read-only tables – expected)
```

T9 — Pre-Restore Auto-Snapshot

```
After restore: backup list contains new entry:
[pre-restore] pre-restore-1779923917345-1trbz
Note: "Auto-snapshot before restoring backup: manual-1779923891917-6m1ld"
Pre-restore snapshot auto-created before any restore ✓
```

T10 — Delete

```
DELETE /api/admin/backup/:id → {"ok":true}
(Note: API allows deleting pre-restore entries; UI disables the button for protection)
```

T11 — Daily Auto-Backup Scheduler

```
Fired ~1 min after server startup (as designed)
daily-1779923913546-7q3rr created at 23:18:33 ✓
Server started at ~23:17
```

T12 — Weekly Auto-Backup Scheduler

```
Fired ~2 min after server startup (as designed)
weekly-1779923973565-v9aiv created at 23:19:33 ✓
Both schedulers confirmed running – no cron dependency needed
```

Notes / Observations

1. **tablesRestored: 16** — 4 tables not restored. This is correct: **backups** table itself is excluded from restore (can't overwrite backup state mid-restore), plus any system-level tables.
 2. **Pre-restore delete via API** — API allows deleting pre-restore entries. Protection is UI-only (button disabled). If API-level protection is desired, a check can be added to the DELETE handler.
 3. **Weekly scheduler** fired at exactly 2min mark — confirms interval timing is accurate.
 4. **48,690 bytes** per backup snapshot — lean and efficient for full DB state.
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Retention Policy (implemented, not yet testable in dev)

- Manual: unlimited
- Daily: keep last 7
- Weekly: keep last 4
- Pre-restore: unlimited (auto-created, UI-protected)